

## 16. Compare Different Classification Algorithms

### Decision Tree and KNN

```
from sklearn.model_selection import train_test_split
```

```
from sklearn.tree import DecisionTreeClassifier
```

```
from sklearn.neighbors import KNeighborsClassifier
```

```
from sklearn.metrics import accuracy_score
```

```
X = [[9.2,90,88],[8.8,85,82],[8.5,80,79],  
      [7.2,72,70],[6.8,68,65],[9.0,92,90],  
      [7.5,75,74],[8.6,84,81],[6.5,60,58],  
      [8.9,88,86]]
```

```
y = ["Placed","Placed","Placed","Not Placed","Not Placed",  
      "Placed","Not Placed","Placed","Not Placed","Placed"]
```

```
X_train,X_test,y_train,y_test = train_test_split(  
    X,y,test_size=0.3,random_state=42)
```

```
dt = DecisionTreeClassifier(random_state=42)
```

```
knn = KNeighborsClassifier(n_neighbors=3)
```

```
dt.fit(X_train,y_train)
```

```
knn.fit(X_train,y_train)
```

```
p1 = dt.predict(X_test)
```

```
p2 = knn.predict(X_test)
```

```
print("Decision Tree Accuracy:", accuracy_score(y_test,p1))
```

```
print("KNN Accuracy:", accuracy_score(y_test,p2))
```

## OUTPUT

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```
Python 3.14.6 (tags/v3.14.6:c63aec6, Jun 10 2026, 10:26:10) [MSC v.1944 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.

= RESTART: C:/Users/yuvas/Downloads/win-g2010-1_3-n_mcd/win/LIB/machine learning lab/ML LAB CODE.py
Decision Tree Accuracy: 1.0
KNN Accuracy: 1.0
|
```